King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The four vertices of a parallelogram $ABCD$ taken in order are $A(2,9,1)$, $B(3,11,4)$, $C(1,12,a)$ and $D(0,b,2)$. Then $a+b=$

(a) 15

(b) 5

(c) 6

(d) 20

(e) -15

2. The area of the region interior to $r = 2+\sin\theta$ and below the polar axis is

(a) $\dfrac{9\pi}{4}-4$

(b) $\dfrac{9\pi}{2}-4$

(c) $\dfrac{9\pi}{4}$

(d) $\dfrac{9\pi}{4}-8$

(e) $\dfrac{9\pi}{2}$

3. The vector component of $\vec{u}=\langle 5,-1,-1\rangle$ orthogonal to $\vec{v}=\langle -1,5,8\rangle$ is

(a) $\vec{u}+\dfrac{6}{\sqrt{10}}\vec{v}$

(b) $\vec{u}+\dfrac{1}{5}\vec{v}$

(c) $\vec{0}$

(d) $\vec{u}-\dfrac{1}{5}\vec{v}$

(e) $-\vec{u}+\dfrac{1}{5}\vec{v}$

4. Let $\vec{v} = \langle a, b\rangle$ be the vector of magnitude 5 and in the same direction as the vector $\langle -1, 2\rangle$. Then $a + b =$

(a) $-\sqrt{5}$

(b) $3\sqrt{5}$

(c) $\sqrt{5}$

(d) 5

(e) $2\sqrt{5}$

5. If the standard equation of the sphere with center $(-7, 7, 6)$ and tangent to the xy-plane is $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$, then $a + b + c + r^2 =$

(a) 26

(b) 14

(c) 12

(d) 6

(e) 42

6. The curve represented by the parametric equations

$$x = \frac{1}{\sqrt{t+1}} \quad \text{and} \quad y = \frac{t}{t+1}; \qquad -1 < t \leq 0$$

is given by the rectangular equation

(a) $y = 1 - x^2, \quad 0 \leq x \leq 1$

(b) $y = \dfrac{x}{1+x}, \quad -1 < x \leq 0$

(c) $y = x^2 - 1, \quad x > 0$

(d) $y = -1 - x^2, \quad 0 < x < 1$

(e) $y = 1 - x^2, \quad x \geq 1$

7. The slope of the tangent line to the graph of $r = 2 + 3\sin\theta$ at $\theta = \pi$ is

(a) $\dfrac{1}{2}$

(b) $-\dfrac{2}{3}$

(c) $-\dfrac{1}{2}$

(d) 0

(e) $\dfrac{2}{3}$

8. An interval of θ over which the rose curve represented by the polar equation $r = \sin 3\theta$ is traced only once is

(a) $[0, 2\pi)$

(b) $[0, \pi)$

(c) $\left[0, \frac{2\pi}{3}\right)$

(d) $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right)$

(e) $\left[\frac{\pi}{3}, \pi\right)$

9. The area of the triangle with vertices $A(2, -3, 4)$, $B(0, 1, 2)$ and $C(-1, 2, 0)$ is

(a) $\sqrt{7}$

(b) $\sqrt{13}$

(c) $\sqrt{5}$

(d) $\sqrt{3}$

(e) $\sqrt{11}$

10. The area of the region that lies inside $r = 1 + \cos\theta$ and outside $r = \cos\theta$ is

(a) $\frac{5\pi}{4}$

(b) $\frac{\pi}{3}$

(c) $\frac{5\pi}{2}$

(d) $\frac{7\pi}{4}$

(e) π

11. A possible set of parametric equations for the line through $(0, 0)$ and $(4, -7)$ is

(a) $x = 4 - 4t, \quad y = 7 - 7t$

(b) $x = 1 - 4t, \quad y = 1 + 7t$

(c) $x = -7t, \quad y = 4t$

(d) $x = 4 + 4t, \quad y = -7 - 7t$

(e) $x = -4 + 4t, \quad y = -7t$

12. The parametric curve given by

$$x = 2t + \ln t, \quad \text{and} \quad y = 2t - \ln t$$

is

(a) concave upward when $t > 0$

(b) concave downward when $0 < t < \dfrac{1}{4}$

(c) concave upward when $0 < t < \dfrac{1}{2}$ and downward when $t > \dfrac{1}{2}$

(d) concave downward when $t > 0$

(e) concave downward when $0 < t < \dfrac{1}{2}$ and upward when $t > \dfrac{1}{2}$

13. The triangle with vertices $(-3, 0, 0)$, $(0, 0, 0)$ and $(1, 2, 3)$

(a) is acute

(b) is obtuse

(c) is equilateral

(d) is right

(e) has zero area

14. Consider the three vectors $\vec{u} = \langle 1, 3, 1\rangle$, $\vec{v} = \langle 0, 6, 6\rangle$ and $\vec{w} = \langle -4, 0, -4\rangle$. Then $\vec{u} \cdot (\vec{v} \times \vec{w}) =$

(a) 64

(b) -72

(c) 72

(d) -64

(e) 24

15. Let C be the curve given by the parametric equations

$$x = \cos\theta, \quad y = 2\sin 2\theta \quad \text{on } [0, 2\pi).$$

Let m be the number of points on the curve at which the tangent line is horizontal, and n be the number of points on the curve at which the tangent line is vertical. Then

(a) $m = 0$ and $n = 4$

(b) $m = 4$ and $n = 2$

(c) $m = 4$ and $n = 0$

(d) $m = 2$ and $n = 4$

(e) $m = 2$ and $n = 0$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\overrightarrow{AB} = \overrightarrow{DC}$: $\langle 1,2,3\rangle = \langle 1, 12-b, a-2\rangle$, so $a=5$, $b=10$.
2	**(a)**	$A = \frac{1}{2}\int_{\pi}^{2\pi}(2+\sin\theta)^2 d\theta = \frac{9\pi}{4} - 4$.
3	**(b)**	proj $= \frac{-18}{90}\vec{v} = -\frac{1}{5}\vec{v}$, so the orthogonal part is $\vec{u} + \frac{1}{5}\vec{v}$.
4	**(c)**	$\vec{v} = \sqrt{5}\langle -1, 2\rangle$, so $a+b = -\sqrt{5} + 2\sqrt{5} = \sqrt{5}$.
5	**(e)**	$(a,b,c) = (-7,7,6)$, $r=6$: $-7+7+6+36 = 42$.
6	**(e)**	$x^2 = \frac{1}{t+1}$, $y = 1 - \frac{1}{t+1} = 1 - x^2$; $0 < t+1 \le 1$ gives $x \ge 1$.
7	**(b)**	$r=2$, $r' = -3$ at π: $\frac{dy}{dx} = \frac{-2}{3} = -\frac{2}{3}$.
8	**(b)**	$n=3$ is odd: $(r, \theta+\pi)$ repeats (r,θ), so $[0,\pi)$ traces it once.
9	**(e)**	$\overrightarrow{AB} \times \overrightarrow{AC} = \langle -6,-2,2\rangle$, area $= \frac{1}{2}\sqrt{44} = \sqrt{11}$.
10	**(a)**	Circle lies inside cardioid: $\frac{3\pi}{2} - \frac{\pi}{4} = \frac{5\pi}{4}$.
11	**(d)**	Direction $\langle 4,-7\rangle$; $x = 4+4t$, $y = -7-7t$ lies on $7x+4y=0$ with that direction.
12	**(a)**	$\frac{dy}{dx} = \frac{2t-1}{2t+1}$, $\frac{d^2y}{dx^2} = \frac{4t}{(2t+1)^3} > 0$ for $t>0$.
13	**(b)**	At the origin, $\langle -3,0,0\rangle \cdot \langle 1,2,3\rangle = -3 < 0$: obtuse angle.
14	**(b)**	$\det\begin{bmatrix} 1 & 3 & 1 \\ 0 & 6 & 6 \\ -4 & 0 & -4 \end{bmatrix} = -24 - 72 + 24 = -72$.
15	**(b)**	$dy/d\theta = 4\cos 2\theta = 0$: 4 points; $dx/d\theta = -\sin\theta = 0$: 2 points.